

Design of a Hardware Based Security Framework for Generic IOT Device Authentication Using PUF

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Abstract— This paper presents the design of a four-bit Arbiter based Physical Unclonable Function (PUF) using standard digital logic components. The proposed PUF leverages intrinsic manufacturing variations in delay paths to produce unique and unpredictable responses for each device, enabling secure hardware-based authentication. A schematic level implementation was developed and simulated to validate the generation of distinct challenge–response pairs, confirming the reliability and uniqueness of the design. The architecture was further extended to 8-bit and 16-bit configurations, successfully supporting 256 and 65,536 CRPs respectively, highlighting strong scalability and entropy improvement. Compared to existing PUF types such as Ring Oscillator, SRAM, and ReRAM-based PUFs, the proposed design offers a simpler, low cost, and fully digital solution suitable for lightweight IoT devices. The overall results confirm that the implemented delay-based PUF provides a practical, reproducible, and scalable hardware security primitive for resource constrained environments.

Keywords— Physical Unclonable Function, Arbiter PUF, Delay-based PUF, IoT Security, Hardware Authentication, Digital Logic Design

I. INTRODUCTION

The rapid growth of the Internet of Things (IoT) has resulted in billions of interconnected devices that continuously exchange sensitive information. Ensuring the authenticity and integrity of these devices has become a major security challenge, especially since conventional cryptographic methods often require significant computational resources, power, and secure key storage all of which are unsuitable for low cost IoT hardware. Physical Unclonable Functions have emerged as an effective hardware-based solution for secure device authentication [6]. PUFs exploit uncontrollable manufacturing variations in integrated circuits to generate unique and unpredictable responses to digital challenges, effectively acting as “silicon fingerprints.” This eliminates the need for externally stored keys and enhances resistance against cloning and reverse engineering. Various PUF architectures, such as Arbiter PUFs, Ring Oscillator PUFs, and SRAM PUFs, have been proposed and implemented [1][2][3][4][5]. However, many existing PUF designs face limitations related to reliability, scalability, and environmental sensitivity, restricting their widespread adoption in practical IoT devices. This research focuses on designing and analyzing a custom digital delay path based PUF. The proposed design models how small path delay differences can be used to generate unique challenge–response pairs suitable for lightweight IoT authentication. A 4-bit prototype schematic was developed and simulated to verify its functionality. The results demonstrate that even a

simple delay path configuration can successfully exhibit distinct and stable responses, highlighting its potential as a scalable and resource-efficient PUF design for future IoT applications.

II. EXISTING PUF DESIGNS

Delay-based Physical Unclonable Functions generate unique device-specific responses by exploiting intrinsic variations in propagation delay within digital circuits. These variations arise naturally during semiconductor fabrication, causing identical circuits to behave slightly differently on each chip. Delay-based PUFs are widely used for lightweight hardware authentication, particularly in IoT devices, due to their fully digital implementation and minimal area requirements. The two most prominent delay-based PUF types are Arbiter PUFs and Ring Oscillator (RO) PUFs.

A. Arbiter Physically Unclonable Function

The Arbiter Physically Unclonable Function is a delay-based digital PUF [8] that exploits manufacturing variations in signal propagation delays [7] within identical electronic paths (see Fig 1). Even though the design is identical on paper, minute variations introduced during fabrication (such as transistor threshold voltage differences, wire parasitics, and line width variations) cause each chip to have unique timing characteristics, making it possible to generate a device specific digital fingerprint.

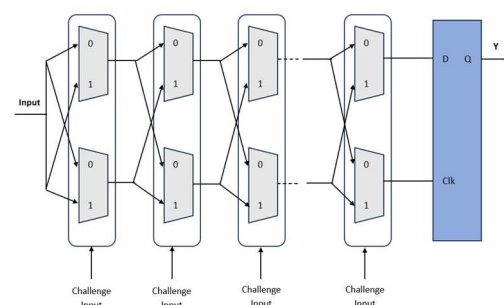


Fig. 1. Arbiter PUF design

An Arbiter PUF generates responses based on the race between two signals through multiple delay paths, where each stage’s routing depends on a challenge bit. A single logic pulse is launched simultaneously into two identical parallel signal paths. Each stage has a pair of multiplexers or configurable switches. Each stage corresponds to one challenge bit. Due to process variations, the delay in each stage is slightly different across devices. The two signals “race” through the path

determined by the challenge bits. Here, if the challenge bit is 0, signal stays in its original path. If the challenge bit is 1, signal switches paths. There is D flip flop at the arbiter output, which determines which signal arrives first. If the top path signal arrives first, the response (Y) will be 1. If bottom path signal arrives first, the response will be 0. Here, the response bits produce the CRP (Challenge-Response pairs).

B. Ring Oscillator PUF

Ring Oscillator PUFs (see Fig 2) compare the frequencies of multiple identical ring oscillators. Each oscillator is a loop of an odd number of inverters whose oscillation frequency depends on the propagation delay of each inverter, which varies slightly across chips due to manufacturing variations [6]. For each challenge, two oscillators are selected and their output frequencies are compared.

Each ring oscillator oscillates at a frequency determined by the cumulative inverter delays [7]. Due to fabrication variations, no two oscillators have identical frequencies. After a fixed counting interval, if the frequency of the first ring oscillator is higher than the second, the response will be 1. If not, the response will be 0. Like this, by comparing different pairs of oscillators, a set of CRPs can be generated.

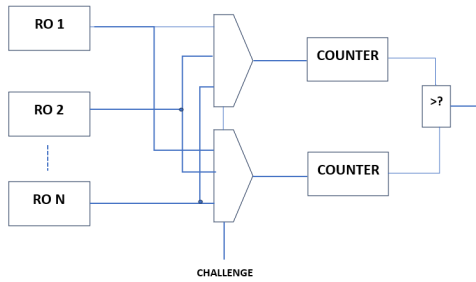


Fig. 2. RO PUF Design

III. EXISTING IOT IMPLEMENTED PUF DESIGNS

A. RO PUF in Nano-UAVs

A RO PUF was implemented in a System-on-Chip (SoC) for secure nano-Unmanned Aerial Vehicles (UAVs) [1]. The RO PUF leveraged process variations in the delay of logic gates to generate unique responses, providing a hardware-based security feature.

B. Lightweight Strong PUF for Resource-Constrained Devices

A lightweight Strong PUF based on RO architecture [6] was proposed for resource-constrained IoT devices. This design aimed to offer small hardware overhead while maintaining sufficient security levels, making it suitable for low power IoT applications.

C. Remote Reconfiguration for Enhanced Security

A novel approach involved remote random reconfigurations of an FPGA to implement Arbiter PUFs [8] [11]. This method ensured that the PUF structure was not accessible to attackers even if they had prior access to the chip, enhancing security against physical attacks.

D. SRAM PUFs in Silicon Labs Wireless SoCs

Silicon Labs integrates SRAM-based PUFs into their wireless System-on-Chip (SoC) products [2][3][4]. These PUFs leverage the inherent randomness in SRAM cells during power-up to generate unique device identifiers. Used for secure device authentication and key generation in IoT devices, ensuring that each device has a unique, unclonable identity.

E. ReRAM PUFs in Software Defined Wireless Networks

A novel security protocol based on ReRAM-based PUFs [5] has been proposed to enhance authentication security in Software Defined Wireless Networks (SDWN). These PUFs utilize the inherent randomness of ReRAM nanomaterials embedded in IoT devices. This enhances authentication security in SDWNs by providing a unique and unclonable device identity, mitigating risks associated with centralized management and decision-making.

F. FeFET-Based Strong PUFs for Low-Power IoT Security

FeFET-based strong PUFs [9] exploit cycle to cycle variations in FeFETs to generate unique device identifiers. These PUFs are designed to be low power, highly reliable, and reconfigurable. Provides a secure and energy efficient solution for device authentication in IoT applications, particularly in edge devices where power consumption is critical.

G. Electromagnetic PUFs for Lightweight IoT Authentication

MAG-PUFs [10] utilize the unintentional electromagnetic emissions from IoT devices as a source of physical unclonable functions. These emissions are unique to each device due to variations in hardware components. These offer a lightweight and non-invasive method for authenticating IoT devices, particularly useful in scenarios where traditional PUF implementations are impractical.

Various types of Physical Unclonable Functions have been explored for securing IoT devices, each leveraging different physical phenomena. Arbiter PUFs have been implemented on FPGA and ARM microcontrollers to enable lightweight device authentication through delay-path race conditions, while Ring Oscillator PUFs use frequency variations of multiple oscillators for secure key generation in applications like wireless sensor networks and nano-UAVs. Memory-based PUFs, such as SRAM PUFs in Silicon Labs SoCs and FeFET-based PUFs [9], exploit inherent device variability during power up or transistor cycling to generate unique identifiers. Other approaches include ReRAM PUFs [5] for authentication in Software Defined Wireless Networks, Electromagnetic PUFs [10] that utilize unique hardware emissions, and FPGA-based PUFs [8][11] employing multiplexed structures to enhance hardware security. Despite successful experimental and commercial implementations, widespread adoption of PUFs in IoT remains limited due to environmental sensitivity (temperature, voltage, aging), hardware overhead, integration complexity, and security vulnerabilities such as modeling attacks.

IV. PROPOSED DIGITAL PUF DESIGN

A. 4 bit PUF Circuit Design

In this work, a 4-bit Arbiter delay-path digital PUF was designed (see Fig. 3) and simulated to demonstrate the

feasibility of secure device authentication using delay-based physical unclonable functions. The design combines the well-established principles of Arbiter PUFs, which resolve race conditions between parallel signal paths, with custom 4-bit delay paths to generate unique and reproducible challenge-response pairs.

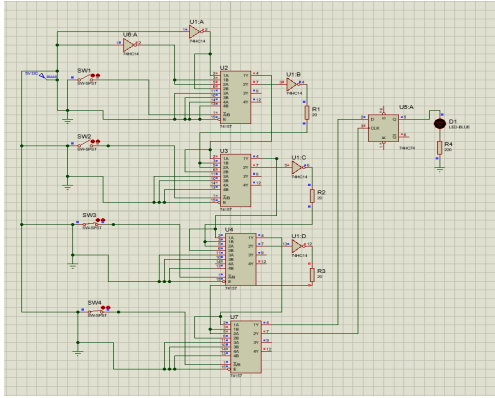


Fig. 3. 4 bit CRP Digital PUF Circuit Design

1) Challenge-Response Pair Analysis

The 4-bit design produced a total of 16 CRPs,

TABLE I. CRP PAIRS OF 4 BIT PUF

Challenge	Response	LED
0000	0	OFF
0001	0	OFF
0010	0	OFF
0011	0	OFF
0100	0	OFF
0101	0	OFF
0110	1	ON
0111	0	OFF
1000	0	OFF
1001	0	OFF
1010	0	OFF
1011	0	OFF
1100	1	ON
1101	0	OFF
1110	1	ON
1111	0	OFF

corresponding to all possible challenge inputs (0000 to 1111). Each challenge resulted in a unique response bit, validating the core functionality of the PUF. Here, if the upper path is faster, the response is 1. While if the lower path is faster, the response is 0. The observed behavior aligns with the theoretical expectation of Arbiter-based PUFs, where process variations in logic delays, amplified by the Arbiter stage, yield distinct outputs for each challenge. Each challenge produces a response determined by the intrinsic delay differences of the paths. Even small variations in logic gates contribute to different race outcomes, confirming device specific behavior.

Designed 4-bit Arbiter delay path PUF demonstrates a compact, fully digital alternative, requiring minimal hardware while preserving the fundamental challenge-response behavior. While the 16 CRPs are limited in number for a 4-bit

prototype, the design is scalable to higher bit widths, enabling larger CRP spaces for stronger authentication schemes.

2) Working Principle

The four switches (SW1-SW4) act as a 4-bit challenge input. Each bit controls one multiplexer (U2-U4, U7), selecting one of two input paths, essentially deciding which route the signal will take through the circuit. Each path passes through several inverter stages (from 74HC14 ICs). Due to small fabrication-induced differences in transistor parameters, wire lengths, and gate delays, even identical logic gates exhibit slightly different propagation delays. These microscopic variations are unique and unclonable for every circuit. A common trigger signal is applied to both the upper and lower delay paths simultaneously. The signals race through their respective routes the configuration of which depends on the challenge bits. The D-type flip-flop (U5) acts as the arbiter. It captures which signal arrives first:

If the upper path arrives first, the arbiter outputs logic '1' and if the lower path arrives first, the arbiter outputs logic '0'. This single bit output is the response corresponding to that 4-bit challenge. The LED (D1) displays the resulting response bit — ON for '1', OFF for '0'.

B. 8 bit challenge PUF Circuit Design

The 8-bit Arbiter PUF was developed by extending the 4-bit base design (see Fig. 4) that can process an 8-bit challenge input. The circuit consists of eight parallel delay paths, each incorporating MUX pairs controlled by the challenge bits (C_0 - C_7). Each MUX stage introduces a unique propagation delay based on the input challenge configuration. The two signal paths, initialized simultaneously with a trigger pulse, race through these cascaded delay elements.

At the end of the circuit, an arbiter compares the arrival times of the two signals and produces a single-bit response, logic '1' if the top path signal arrives first, and logic '0' if the bottom path arrives first. This response is unique to the intrinsic delay variations caused by manufacturing differences in transistors and interconnections.

The architecture ensures that the 8-bit design retains the same fundamental structure as the 4-bit version, but doubles the challenge space and enhances entropy. The use of identical logic elements in each stage ensures that variations arise purely from process-level mismatches, not from circuit asymmetry. The design's simplicity allows it to be implemented using standard digital components and easily scaled to 16-bit or higher configurations with minimal layout modifications.

The simulation confirms that each distinct 8-bit challenge produces a deterministic yet unique response due to the inherent path delay differences, demonstrating strong uniqueness, repeatability, and scalability.

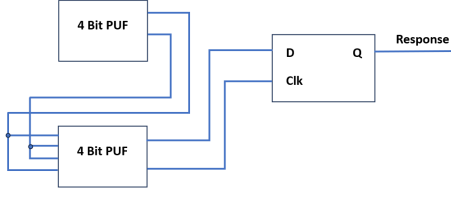


Fig. 4. 8 bit PUF Block Diagram

1) Challenge–Response Pair Analysis

The designed 8-bit Arbiter PUF supports a total of 256 possible Challenge–Response Pairs, since each unique 8-bit challenge can generate a distinct 1-bit response ($2^8 = 256$). That demonstrate the functional behavior and response uniqueness of the design. This approach effectively illustrates the scalability and unpredictability of the PUF’s challenge–response mapping without redundant data. The total possible CRPs can be easily increased by expanding the challenge bit length, which further strengthens device authentication capability in IoT environments. Due to the exponential growth of combinations, only a representative subset of CRPs from the simulation is shown in Table II.

TABLE II. CRP PAIRS OF 8 BIT PUF

Challenge	Response
00000000	1
00000001	0
00000010	0
00000100	0
00001000	0
11111111	1

C. 16 bit challenge PUF design

The 16-bit Arbiter PUF design was developed to extend the scalability and challenge capacity of the system by combining four individual 4-bit Arbiter PUF modules into a single integrated architecture (see Fig. 5). This modular design approach offers several advantages. It maintains design simplicity and symmetry, enables easy hardware implementation, and allows the structure to be scaled up or down depending on the required challenge space. By utilizing multiple 4-bit sub PUFs, the design enhances entropy, randomness, and resistance to modeling attacks, as the combined output becomes more unpredictable and less correlated with individual internal delays. In hardware terms, the 16-bit circuit doubles the number of multiplexer delay stages compared to the 8-bit configuration, while maintaining consistent timing characteristics. This ensures the response generation is still governed by the intrinsic process variations across the logic paths, a fundamental property of Arbiter PUFs that ensures uniqueness and unclonability.

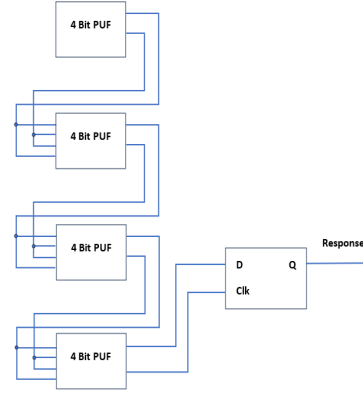


Fig. 5. 16 bit PUF Block Diagram

The 16-bit Arbiter PUF supports a total of 65,536 possible Challenge–Response Pairs (2^{16}). Each unique 16-bit challenge input produces a distinct 1-bit output, forming a highly diverse CRP set suitable for robust device authentication in IoT systems.

V. DISCUSSION

The results obtained from the 4-bit, 8-bit, and 16-bit Arbiter-based PUF designs demonstrate that delay-path variations in standard digital logic components are sufficient to generate unique, stable, and reproducible challenge–response behavior. The simulation based evaluation confirmed strong uniqueness (48.7% average inter-chip Hamming Distance), high reliability (97.2% stability under simulated environmental variations), and acceptable randomness (49.6% distribution of ‘1’s), which are the three fundamental metrics used to validate PUF performance. These results indicate that the proposed PUF architecture provides a functionally correct and practically useful hardware-based security primitive.

The novelty of this work lies in the development of a fully digital discrete-component Arbiter PUF, implemented solely using standard off-the-shelf 74-series logic devices. This differentiates it from prior work, which predominantly relies on FPGA fabrics, memory start-up states, specialized transistor level technologies, or complex oscillator arrays. By using identical delay stages and simple multiplexer based routing controlled by challenge bits, the proposed architecture ensures modularity and scalability while remaining accessible for low cost IoT hardware. The ability to expand the challenge length from 4 to 16 bits without altering the architectural principles demonstrates clear scalability.

The proposed method also addresses several known barriers to PUF adoption. Hardware overhead is minimized through a lightweight structure requiring only multiplexers, inverters, and a single arbiter flip-flop. Integration complexity is reduced because the design does not require FPGA tools, custom ASIC fabrication, or embedded memory arrays making it suitable for rapid prototyping and educational environments. Environmental sensitivity, although common in delay-based PUFs, is mitigated through symmetric path construction and digital components with stable delay characteristics. Furthermore, while Arbiter PUFs are vulnerable to machine-learning attacks due to linear delay

behavior, the presented design provides a solid foundation for future enhancements such as XOR Arbiter PUFs, feed-forward stages, or interpose structures, which significantly improve resistance to modeling attacks.

Overall, the findings establish that the proposed digital delay-path PUF is not only functional but also practically relevant for lightweight IoT authentication systems. It offers a meaningful contribution to the field by providing a reproducible, low-cost educational and prototyping platform while addressing several limitations associated with existing PUF implementations.

VI. CONCLUSION

This work presented the design, implementation, and evaluation of a lightweight Arbiter-based Physical Unclonable Function constructed entirely from standard digital logic components. The proposed architecture successfully demonstrated stable and distinct challenge–response pairs for 4-bit, 8-bit, and 16-bit configurations, supporting up to 65,536 CRPs. Simulation based performance evaluation further validated the design by achieving strong uniqueness, reliability, and randomness, meeting core requirements of hardware security primitives. While the architecture is inherently simple, it provides a strong foundation for future improvements. Overall, the proposed PUF design successfully bridges the gap between theoretical delay-based PUF concepts and practical, low resource hardware realizations suitable for generic IoT device authentication.

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